

Project Design Phase-II Technology Stack (Architecture & Stack)

Date	03 October 2026
Team ID	To be provided
Project Name	Script-Controlled ACL – Restrict Record Access Based on Field Value
Maximum Marks	4 Marks

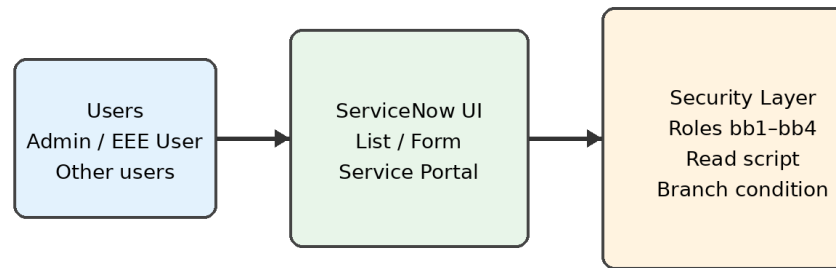
Technical Architecture:

The solution is implemented inside a ServiceNow-managed platform. Users access list, form, and optionally Service Portal interfaces; ServiceNow evaluates roles and ACL scripts before exposing records or operations.

Project architecture: identity and role layer → ServiceNow UI → ACL/security layer → u_institution_details data layer.

Reference: ServiceNow platform configuration and ACL documentation; project-specific implementation details are based on the supplied lab file. <https://developer.ibm.com/patterns/ai-powered-backend-system-for-order-processing-during-pandemics/>

ServiceNow ACL Solution Architecture



Guidelines:

- Include all the processes (As an application logic / Technology Block)
- Provide infrastructural demarcation (Local / Cloud)
- Indicate external interfaces (third party API's etc.)
- Indicate Data Storage components / services
- Indicate interface to machine learning models (if applicable)

Deployment: ServiceNow-managed instance; no separate Node.js/MongoDB front

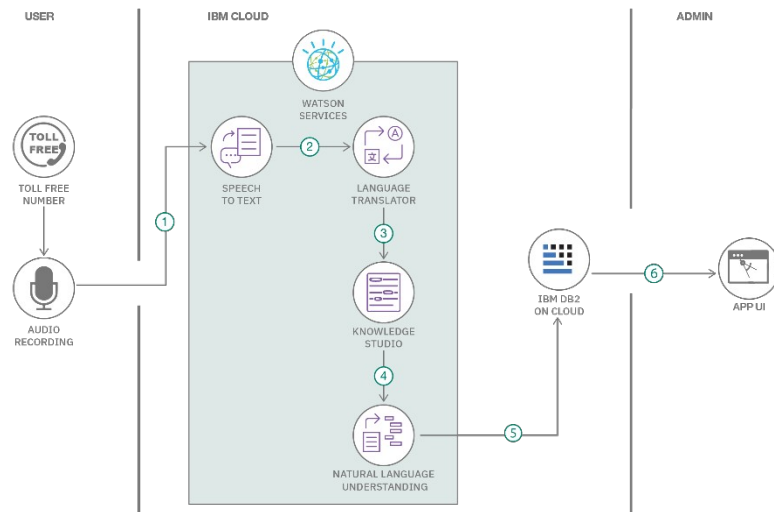


Table-1 : Components & Technologies:

S.No	Component	Description	Technology
1.	User Interface	Users interact through ServiceNow list, form, and supported Service Portal views.	ServiceNow UI / Service Portal
2.	Application Logic-1	Evaluates Read access using administrator override, bb1 role, and Branch = EEE condition.	ServiceNow Advanced Record ACL / GlideSystem
3.	Application Logic-2	Controls record creation using the bb2 role.	ServiceNow Create ACL
4.	Application Logic-3	Controls updates and deletion using bb3 and bb4 roles.	ServiceNow Write and Delete ACLs
5.	Database	Stores u_institution_details records and fields including Branch, references, contact data, and Description.	ServiceNow table storage
6.	Cloud Database	The supplied project does not use a separately managed cloud database.	ServiceNow-managed data layer
7.	File Storage	No project file storage is required; screenshots/evidence are maintained separately for testing.	ServiceNow attachments or approved project storage
8.	External API-1	No external API is specified in the supplied project.	Not applicable
9.	External API-2	No external API is specified in the supplied project.	Not applicable
10.	Machine Learning Model	No machine-learning model is used; authorization is deterministic.	Not applicable
11.	Infrastructure (Server / Cloud)	Application runs on the organization's ServiceNow instance; administration requires authorized roles and security_admin elevation.	ServiceNow cloud/managed instance

Table-2: Application Characteristics:

S.No	Characteristics	Description	Technology
1.	Open-Source Frameworks	No external application framework is used in the supplied project.	ServiceNow platform
2.	Security Implementations	Record ACLs, role-based access, advanced script, Branch data condition, security_admin elevation, and least-privilege operation	ServiceNow ACL / GlideSystem

S.No	Characteristics	Description	Technology
		roles.	
3.	Scalable Architecture	Centralized table ACLs evaluate access consistently as records and users grow; add regression tests and role governance for expansion.	ServiceNow platform controls
4.	Availability	Availability depends on the ServiceNow instance and its configured authentication, platform, and recovery services.	ServiceNow managed availability
5.	Performance	Measure list/form response, filtered records, controlled concurrency, and sustained sessions; avoid unnecessary ACL complexity and validate representative data volumes.	ACL evaluation and platform monitoring

References:

<https://c4model.com/>ServiceNow product documentation: <https://www.servicenow.com/docs/>

<https://developer.ibm.com/patterns/online-order-processing-system-during-pandemic/>Project source document: Script-Controlled ACL – Restrict Record Access Based on Field Value

<https://www.ibm.com/cloud/architecture>Instance-specific ACL configuration and test evidence: To be attached by project team

<https://aws.amazon.com/architecture>

<https://medium.com/the-internal-startup/how-to-draw-useful-technical-architecture-diagrams-2d20c9fda90d>